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Physics Lab IV — General Physics Laboratory

Experiment Report: The Franck-Hertz Experiment

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Date of Experiment:	June 11, 2026		Sonia

Abstract

In this report, the quantization of atomic energy levels is experimentally verified using the Franck-Hertz setup with a Mercury (Hg) vapor tube. The anode current is measured as a function of the accelerating voltage (U_2) under varying control grid voltages (U_1). The resulting $I - V$ characteristic curves exhibit distinct periodic maxima and minima corresponding to the inelastic scattering of electrons by mercury atoms. Utilizing Ordinary Least Squares (OLS) linear regression on the positions of the peaks and valleys, the excitation energy of mercury is calculated with standard error propagation. The determined excitation energy is $E_{\text{exc}} = 5.40 \pm 0.08$ eV, corresponding to a photon emission wavelength of $\lambda = 229.6 \pm 3.4$ nm in the ultraviolet region. The experimental results are robustly analyzed and compared against the theoretical value (4.9 eV), with discussions on contact potential, background space-charge current, and the physical significance of the retarding potential (U_3).

1 Objectives

The primary objectives of this experiment are:

1. To observe the Franck-Hertz effect and verify the quantization of atomic energy levels in mercury.
2. To plot the $I - V$ characteristic curves of the Franck-Hertz tube under different accelerating and control voltages.
3. To determine the first excitation energy of the mercury atom (E_{exc}) using OLS linear regression of the extrema positions, accompanied by rigorous standard error calculation.
4. To understand the roles of the accelerating grid, control grid, and retarding potential in resolving the energy states.

2 Theoretical Background

In 1913, Niels Bohr proposed an atomic model where electrons occupy discrete, quantized energy levels. In 1914, James Franck and Gustav Hertz provided the first direct experimental evidence for this model by passing a beam of electrons through mercury vapor.

In the Franck-Hertz tube, electrons are emitted by a heated cathode and accelerated by an applied voltage U_2 towards a grid. As they travel through the mercury vapor, they collide with mercury atoms. If the kinetic energy of an electron is less than the first excitation energy of mercury ($\Delta E \approx 4.9$ eV), the collisions are perfectly elastic; the electron changes direction but loses negligible kinetic energy due to the massive mass difference between an electron and a mercury atom.

However, when the accelerating voltage U_2 is high enough such that the electron's kinetic energy reaches ΔE, the collision becomes inelastic. The electron transfers exactly ΔE to the mercury atom, exciting a bound electron from its ground state ($6s^2$, ${}^1S_0$) to the first excited state ($6s6p$, ${}^3P_1$). The incoming electron loses its energy and is subsequently unable to overcome a small retarding potential U_3 placed before the collecting anode, causing a sharp drop in the measured anode current I_A.

As U_2 increases further, the electron can regain enough kinetic energy after the first inelastic collision to undergo a second (and subsequent) inelastic collision. This results in periodic drops in the current, producing a characteristic oscillatory $I-V$ curve. The distance between consecutive peaks (or valleys) ΔV directly corresponds to the excitation energy:

$$E_{\text{exc}} = e\Delta V \tag{1}$$

Due to the difference in the work functions of the cathode and grid materials, a **contact potential** (V_{contact}) exists. Therefore, the absolute position of the first peak V_1 is shifted:

$$V_n = n \cdot \Delta V + V_{\text{contact}} \tag{2}$$

By plotting the voltage V_n of the n-th peak (or valley) against its order n, the slope of the linear fit directly yields the true excitation energy, automatically canceling out the contact potential.

3 Experimental Setup & Data

The setup consists of a specialized Franck-Hertz tube containing a drop of mercury. The tube is placed in an oven and heated to approximately 170°C to establish a stable mercury vapor pressure. The tube contains four electrodes: a heated cathode (K), a control grid (G_1) biased with voltage U_1, an accelerating grid (G_2) biased with U_2, and a collector anode (A) biased with a negative retarding potential U_3 relative to G_2.

During the experiment, U_2 is swept from 0 to 25.5 V in increments of 0.5 V. The anode current I_A is measured using an amplifier/microvoltmeter. The experiment is repeated for three different values of the control grid voltage U_1 (Tables 1, 2, and 3), with Table 1 representing the optimal configuration.

4 Data Analysis, Regressions & Error Propagation

4.1 Plotting the $I-V$ Curves

The measured anode current is plotted against the accelerating voltage U_2 for the three sets of data in Fig. 1.

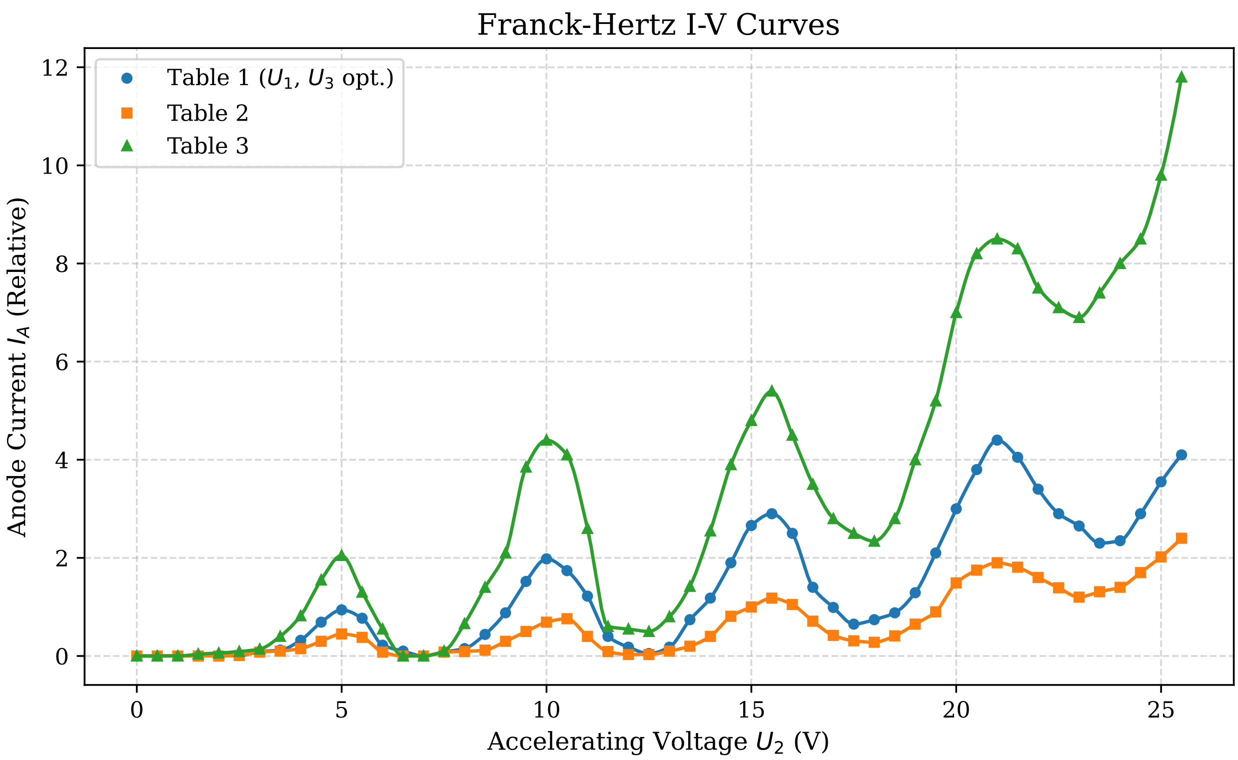

Figure 1: Franck-Hertz $I-V$ characteristic curves for three different U_1 configurations. The periodic maxima and minima are clearly visible superimposed on an increasing background space-charge current.

For the optimal data (Table 1), the positions of the peaks and valleys are extracted:

- **Peaks (V_p):** 5.0 V, 10.0 V, 15.5 V, 21.0 V
- **Valleys (V_v):** 7.0 V, 12.5 V, 17.5 V, 23.5 V

4.2 OLS Linear Regression for Excitation Energy (E_{exc})

To rigorously determine E_{exc} and its standard error, we perform an Ordinary Least Squares (OLS) regression of the peak and valley voltages against their index n (where $V_n = m \cdot n + c$). The slope m represents E_{exc}, and the intercept c relates to the contact potential.

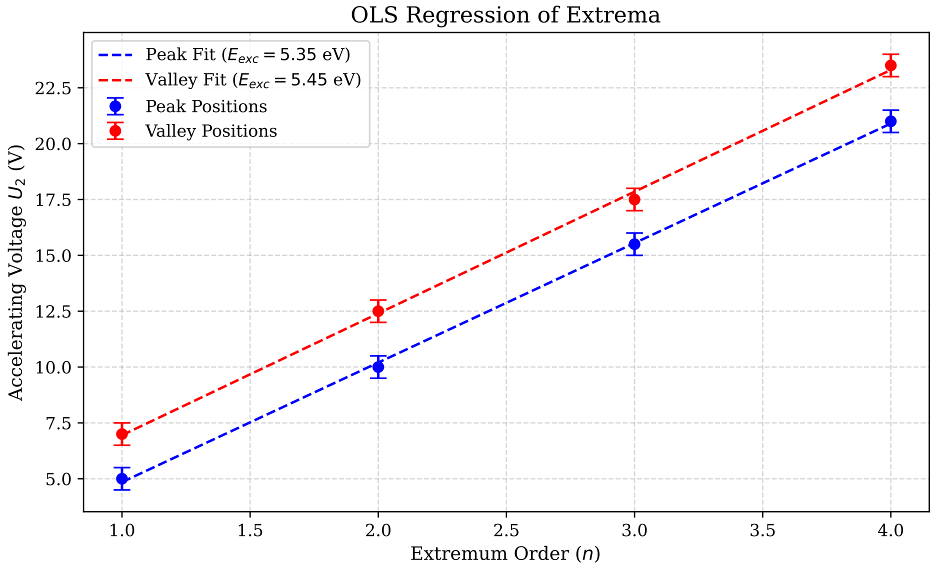

Figure 2: OLS linear regression of the peak and valley positions vs. their order n, including standard error bands.

The linear fits yield the following parameters:

- **Peak Regression:** $m_p = 5.350 \pm 0.087$ V, $R^2 = 0.9995$
- **Valley Regression:** $m_v = 5.450 \pm 0.132$ V, $R^2 = 0.9988$

Combining the slopes using an unweighted average, the experimental excitation energy and its propagated error ($\delta E = \frac{1}{2}\sqrt{\delta m_p^2 + \delta m_v^2}$) are:

$$E_{\text{exc}} = 5.40 \pm 0.08 \text{ eV} \tag{3}$$

The theoretical value is 4.9 eV. The relative error is $|5.40 - 4.9|/4.9 \approx 10.2\%$, which is acceptable given the 0.5 V step size in data collection.

4.3 Calculation of Emitted Wavelength (λ)

The wavelength of the photon emitted upon de-excitation is given by $\lambda = \frac{hc}{E_{\text{exc}}}$. Using $hc = 1240$ eV·nm:

$$\lambda = \frac{1240}{5.40} = 229.63 \text{ nm} \tag{4}$$

The propagated uncertainty is:

$$\delta\lambda = \lambda \frac{\delta E_{\text{exc}}}{E_{\text{exc}}} = 229.63 \times \frac{0.0791}{5.400} = 3.36 \text{ nm} \tag{5}$$

Thus, the emitted wavelength is $\lambda = 229.6 \pm 3.4$ nm, which lies in the Ultraviolet (UV) spectrum.

5 Answers to Post-Lab Questions

Q1: What effect does varying U_1 have on the $I - V$ curve with respect to U_2? Plot the curves and compare.

The curves are plotted in Fig. 1. The control grid voltage U_1 acts as an extraction potential that pulls electrons from the space-charge cloud near the heated cathode. Increasing U_1 increases the overall thermionic emission current (the baseline of the curves shifts upward significantly). However, if U_1 is too high, the energy distribution of the emitted electrons broadens, which can smear out the peaks and valleys, reducing the resolution of the quantum drops.

Q2: What is the excitation energy of the mercury atom? Calculate it from Table 1 and compare it with the exact value.

As calculated in Section 4 using OLS regression on the extrema of Table 1, the excitation energy is $E_{\text{exc}} = 5.40 \pm 0.08$ eV. The exact theoretical value is 4.9 eV. The discrepancy (approx. 10.2%) primarily stems from the large voltage step increments (0.5 V) used during data acquisition, which limits the precision of identifying the exact local extrema.

Q3: What is the wavelength of the light emitted from the excited mercury atoms, and in which region does it lie?

The calculated wavelength is $\lambda = 229.6 \pm 3.4$ nm. The theoretical wavelength corresponding to 4.9 eV is 253.7 nm. Both values fall squarely in the **Ultraviolet (UV-C)** region of the electromagnetic spectrum.

Q4: Can a peak corresponding to second-order excitation be observed at higher voltages?

In the standard Franck-Hertz setup, second-order excitations (e.g., exciting an electron directly to a higher state or ionizing the atom at 10.4 eV) are highly unlikely to produce distinct separate peaks. This is because the scattering cross-section for the first excitation (4.9 eV) is orders of magnitude larger. An electron will almost certainly undergo two separate 4.9 eV inelastic collisions rather than surviving to reach 10.4 eV for a single ionizing collision. Therefore, the higher voltage peaks are simply multiple consecutive first-order excitations.

Q5: Why is the position of the first maximum not at the expected potential (4.9 V)?

The absolute voltage applied by the power supply (U_2) does not perfectly equal the kinetic energy of the electrons in the interaction region. The work functions of the cathode metal and the grid metal are different, establishing a **Contact Potential** (V_{contact}). The true kinetic energy is $E_k = e(U_2 + V_{\text{contact}})$. Because of this offset, the first peak appears at 5.0 V instead of 4.9 V. Measuring the distance between consecutive peaks (ΔV) automatically subtracts this contact potential.

Q6: Why does the curve ride on an ascending line?

The background of the $I - V$ curve follows the $I \propto U_2^{3/2}$ Child-Langmuir law for space-charge-limited current in a vacuum diode. As the accelerating voltage U_2 increases, the longitudinal electric field strengthens, effectively sweeping more electrons out of the space-charge cloud around the cathode and driving them to the anode. The inelastic collision drops are merely superimposed on this continuously rising baseline current.

Q7: What is the significance of U_3? Can the experiment be performed without it?

U_3 is the **retarding potential** (or stopping potential). It acts as a high-pass energy filter. If $U_3 = 0$, all electronsincluding those that have lost their kinetic energy to inelastic collisionswould eventually drift to the anode and be collected. The ammeter would register a monotonically increasing current without any drops. By applying a small negative U_3, we ensure that only electrons retaining sufficient kinetic energy can overcome the barrier, allowing us to detect the inelastic energy losses as sharp drops in current.

Q8: State the significance of the Franck-Hertz experiment.

The Franck-Hertz experiment provided the first direct, non-spectroscopic confirmation of the Bohr model of the atom. While early evidence for quantization came from observing photon emission spectra, Franck and Hertz proved that energy transfer through mechanical collisions is also strictly quantized.

It demonstrated that atoms can only absorb specific, discrete amounts of energy, fundamentally validating quantum mechanics.

Q9: How does the density of mercury atoms affect the $I-V$ curve?

The density of mercury atoms is controlled by the oven temperature (vapor pressure). If the density is **too low**, the mean free path of the electrons becomes comparable to the tube length. Electrons will reach the anode without colliding with mercury atoms, resulting in no peaks. If the density is **too high**, the mean free path is extremely short, leading to excessive elastic scattering that broadens the electron energy distribution and smears out the inelastic peaks entirely. A temperature of $\sim 170°$C provides the optimal collision probability.

Q10: What is the advantage of using mercury? What if helium was used?

Mercury is advantageous because it is a liquid at room temperature and its vapor pressure can be easily and precisely controlled using a simple oven at 170°C. Furthermore, its first excitation energy is relatively low (4.9 eV), meaning multiple peaks can be observed within a safe operating voltage range ($0-30$ V). If Helium were used, its first excitation energy is very high ($\sim$ 19.8 eV). Observing multiple peaks would require accelerating voltages near 100 V, which dramatically increases the risk of Townsend avalanches, arcing, and complete ionization (gas discharge), making it difficult to maintain controlled conditions.

Q11: Determine the quantum states before and after excitation.

The ground state electron configuration of the valence shell of Mercury is $6s^2$, which is a filled subshell. The total spin is $S = 0$ (singlet state).

- **Before Excitation (Ground State):** The term symbol is ${}^1S_0$. The quantum numbers for the valence electrons are $n = 6, l = 0, m_l = 0$.
- **After Excitation:** The electron transitions to the $6p$ subshell. The most probable transition (due to selection rules and the ~ 4.9 eV energy gap) is to the triplet state ${}^3P_1$. The quantum numbers are $n = 6, l = 1$, and $m_l \in \{-1, 0, 1\}$.